Aman Kumar

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RESEARCH AREAS

Visual Analytics, Advanced Analytics, Interactive Machine Learning, Human-Computer Interaction, Explainable AI, Uncertainty, Time Series, Decision Sciences, Large Language Models (LLMs), and Multi-Criteria Decision Support.

EDUCATION

University of Leeds | **September 2023 – December 2024**
Master of Science in Business Analytics and Decision Sciences | *Leeds, United Kingdom*

- Graduated with Distinction; GPA: 4/4.
- Courses: Quantitative Analysis, Advanced Predictive Analytics, Consultancy, Multi-Criteria Decision Analysis, Time Series Forecasting, Advanced Management Decision-Making, Data Visualization, Text Analytics, Machine Learning.

Indian Institute of Science Education and Research Bhopal | **August 2017 – July 2022**
BS–MS Dual Degree in Mathematics | *Bhopal, India*

- Graduated with Distinction; Master's Grade: 10/10.
- Courses: Probability and Statistics, Data Structures & Algorithms, Advanced Linear Algebra, Artificial Intelligence, Machine Learning, Biostatistics, Cryptography, Computational Linguistics, ODEs & PDEs.

RESEARCH EXPERIENCE

Researcher: Personalized Visual Analytics | **March 2025 – Present**
University of Zürich | *Zurich, Switzerland*

- Identified shortcomings in current multi-criteria decision systems that lead to suboptimal, non-transparent rankings.
- Developed an interactive VA framework to capture complex user preferences via explicit inputs and implicit feedback.
- Engineered a prototype with advanced attribute scoring, uncertainty visualization, and LLM-powered natural-language explanations in a human-in-the-loop system.
- Established a scalable, transparent prototype improving decision accuracy and user trust for real-world applications.

Researcher: Evaluating the Stability of SHAP Under Class Imbalance | **January 2024 – October 2024**
University of Leeds | *Leeds, United Kingdom*

- Developed and executed an experimental framework to assess the stability of SHAP values across machine learning models (SVM, KNN, Decision Trees, XGBoost) using an imbalanced diabetic and financial credit fraud dataset.
- Developed SMOTE-Tomek hybrid sampling to create datasets with varying class imbalance ratios (1%, 5%, 10%, till 50%) and introduced a novel metric, Coefficient of Variation (CV), for quantifying SHAP value stability.
- Generated visual insights using SHAP plots and box plots to communicate the impact of class imbalance on feature importance stability, aiding stakeholders in balancing accuracy and interpretability in machine learning models.

Researcher: Fourier Analysis and Roth's Theorem | **July 2021 – July 2022**
Indian Institute of Science Education and Research Bhopal | *Bhopal, India*

- Studied Fourier analytic techniques and applications to Roth's Theorem in additive combinatorics.
- Explored recent advancements in the finite field setting and breakthroughs in upper bounds for the Cap-set problem using the polynomial method.
- Developed understanding of mathematical proofs in additive combinatorics and the elegance of solving problems.

PUBLICATIONS

- **A. Kumar**, M. Tornow, M. Benk, I. Al-Hazwani, and J. Bernard, "Ranking Companion: A Visual Analytics Approach to Item-Based Ranking with Hybrid Item Selection," in *Proc. 30th Int. Conf. Information Visualisation (IV)*, Jul. 2026.
- **A. Kumar**, K. Schaerer, M. Benk, and J. Bernard, "ItForAt: Item-to-Attribute Preference Translation for Interpretable Human-Centered Ranking," *Computers & Graphics*, 2026, *under review.*

STUDENT SUPERVISION

- **Bachelor's Thesis:** Inference of Attribute-Based Ranking Models from Item-Based Preferences
- **Master's Thesis:** Training Item-Based Rankers from Attribute-Based Preferences
- **Master's Project:** Personalized Ranking Using Verbal Preference Elicitation for Multi-Criteria Decision Support

TECHNICAL SKILLS

- **Machine & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Transformers, NLP, Multimodal AI, Predictive Analytics, Feature Engineering, Time Series Forecasting, Ensemble Methods, A/B Testing, AutoML.
- **Programming & Dat—:** Python, R, SQL, TypeScript, Pandas, NumPy, Apache Airflow, Kafka, ETL/ELT Pipelines, FastAPI, Flask, REST APIs, GraphQL, Git & GitHub.
- **Visualization:** Power BI (DAX), Tableau, Matplotlib, Seaborn, Streamlit, React, Next.js, Tailwind CSS.
- **Agentic AI & LLMs:** LLM Agents, Multi-Agent Systems, LangChain, LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, Model Context Protocol (MCP), RAG (Retrieval-Augmented Generation), Agentic RAG, Prompt Engineering
- **Business Frameworks:** SWOT, PESTEL, Root Cause & Cost-Benefit Analysis, KPI Development.

PROJECTS

Machine Learning Model for Predicting ICO Success — **January 2024 – May 2024**

- Developed a model to predict the success of Initial Coin Offerings (ICOs) to assist investors and fundraising teams.
- Analyzed 2,767 ICO records using Logistic Regression, Decision Trees, Random Forest, SVM, and ANN.
- Recommended model selection per stakeholder goal: ANN for investors and Decision Trees for fundraising teams.

Universal Exports Business Operations Analysis — **January 2024 – May 2024**

- Extracted, cleaned, and integrated data from five disparate sources into an interactive Power BI dashboard.
- Computed key metrics (profit margins, monthly sales, transaction counts) via DAX measures; automated financial tracking through visual trends and comparisons.
- Secured £249M in UK sales with a 9.28% increase in air shipments, highlighting a need for sustainability.

Forecasting for PCE — **January 2024 – May 2024**

- Prepared a seasonally adjusted time series dataset of PCE to evaluate trends, volatility, and forecasting methods.
- Built forecasting models (Drift, Holt's Linear, ARIMA) and validated using MAPE and MASE metrics.
- Designed an ARIMA(3,2,2) model with the lowest MAPE (5.30%), delivering precise forecasts for October 2024.

Hotel Review Analysis using Topic Modelling — **January 2024 – May 2024**

- Preprocessed 2,000 reviews and constructed Document-Term Matrices for positive and negative feedback; applied LDA.
- Identified "Service Quality" as the most praised feature (14.41%) and "Food and Dining" as most criticised (14.53%).
- Delivered actionable recommendations to hotel management for strengthening service and addressing weaknesses.

Autonomous Shipment Roll-out Project — **September 2023 – January 2024**

- Analysed performance metrics and allocated resources to maximise delivery efficiency within a £250,000 budget.
- Applied VIKOR and TOPSIS for multi-criteria decision-making across four robot prototypes, balancing cost, reliability, and operational efficiency.
- Selected 29 Deviant units as the optimal configuration, achieving 221 orders/day within 97.72% of budget.

ACADEMIC SERVICE & TEACHING

Peer Reviewer

- IEEE VIS — IEEE Visualization and Visual Analytics Conference
- Information Visualisation — International Journal on Information Visualisation

Teaching Assistant

- ***University of Zürich:*** Interactive Visual Data Analysis
- ***IISER Bhopal:*** Introduction to Programming | Linear Algebra | AI and its Scientific Applications

CONFERENCES AND WORKSHOPS

IV 2026 | *30th International Conference on Information Visualisation, London, United Kingdom* — **July 2026**

EURO Summer School (MCDA/MCDM) | *TU Delft, Netherlands* — **July 2026**

EuroVA 2025 | *16th International EuroVis Workshop on Visual Analytics, Luxembourg* — **June 2025**

IfI Summer School (Human–Computer Interaction) | *University of Zurich, Switzerland* — **June 2025**

Winter School on Deep Learning | *Indian Statistical Institute, India* — **December 2023**

Summer Workshop in Applied Mathematics (SWAM) | *TIFR Bangalore, India* — **July 2021**

LEADERSHIP & AWARDS

- Top 6 in IIM Calcutta Datathon 2.0 for innovative analytics solutions.
- 6^{th} place out of 1,500+ participants in Machine Learning in Agriculture Hackathon.
- Ranked in the top 1% among 1.5M candidates in JEE Advanced.